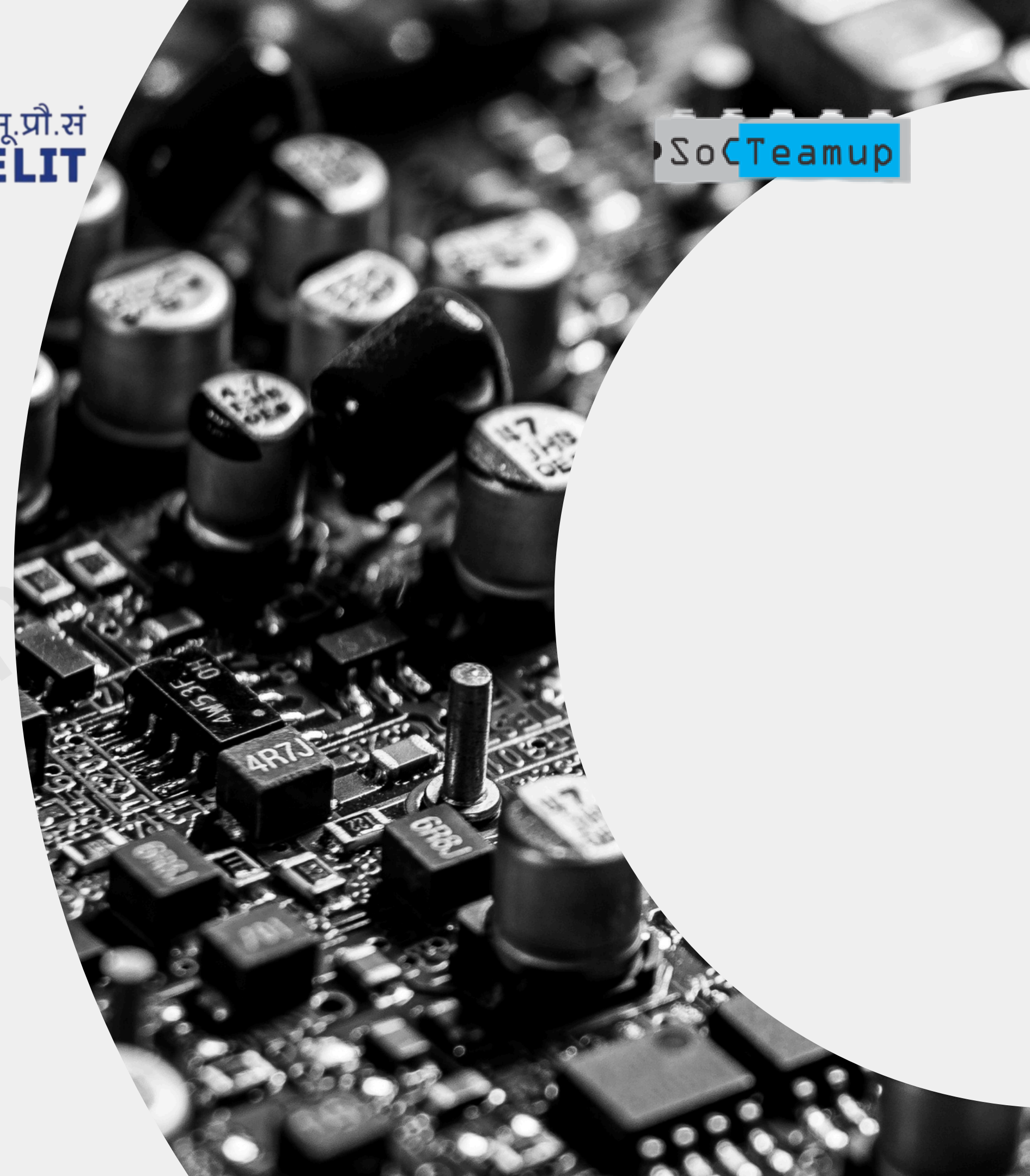


Basics of IR drop and Electromigration (EM) awareness.





Outline

Introduction to Power Integrity
What is IR Drop?
Types of IR Drop (Static and Dynamic)
Factors Influencing IR Drop
Basics of Power Grid Design (PDN)
What is Electromigration (EM)?
Causes and Effects of EM
Relation between IR Drop and EM
Example Analysis Flow (Conceptual)
Best Practices for Mitigation
Summary and Best Practices

➔ Introduction to Power Integrity

- Power integrity ensures that power delivery networks (PDNs) provide stable voltage to every cell in the chip.
- **Two key challenges are:**
 - IR Drop: Voltage loss due to resistance in metal wires.
 - Electromigration (EM): Gradual movement of metal atoms caused by high current density.
- **Poor power integrity leads to:**
 - Timing failures (logic delay)
 - Functional instability
 - Reduced chip lifetime

Goal: Maintain steady voltage and reliable current delivery throughout the layout.

➔ What is IR Drop?

When current flows through a resistive path, voltage drops across that path.

$$\text{IR Drop} = I \times R \text{ (Ohm's Law)}$$

- **Example:** If wire resistance = 0.5Ω and current = 20mA,
Voltage drop = $0.5 \times 0.02 = 10 \text{ mV}$
In a chip, even small voltage drops (tens of mV) can affect timing and logic operation.

➔ Types of IR Drop

Type	Description	Example
Static IR Drop	Caused by constant current through resistive metal lines (depends on R).	High cell utilization regions.
Dynamic IR Drop	Caused by switching transients and simultaneous activity.	Clock edge switching multiple flip-flops.

Static = $I \times R$ (DC drop)

**Dynamic = transient voltage dip due to switching
(AC effect)**

➔ Factors Influencing IR Drop

Wire Resistance

- Thinner or longer wires
- higher R

Current Demand

- More active cells
- higher I

Power Grid Quality

- Sparse power lines
- higher drop

Cell Placement Density

- Clustered switching logic increases local IR drop

Package and bump locations

- Affect global voltage distribution

Term:

PDN (Power Delivery Network): Mesh of power and ground lines connecting all cells to supply pads.

➔ Basics of Power Grid Design

DRC Tool Used:

- The power grid distributes VDD and VSS across the chip using metal layers.
- PDN is built as a mesh or ring structure of power rails.
- Wider metals are used for lower resistance paths.

Metal5: Power ring (VDD/VSS)
Metal4: Horizontal power stripes
Metal3: Vertical power stripes
Via connections between layers

Good PDN Design



Lower IR Drop

➔ IR Drop Example (Conceptual)

Assume:

- Wire resistance = 0.2Ω
- Current = 50mA
- $\rightarrow IR \text{ drop} = 0.2 \times 0.05 = 10\text{mV}$

Code Example (Python):

```
R = 0.2 # Ohms  
I = 0.05 # Amps  
V_drop = R * I  
print(f"IR Drop = {V_drop*1000:.2f} mV")
```

Output:

- IR Drop = 10.00 mV
- Even 10 mV drop may affect timing in sub-1V designs.

What is Electromigration (EM)?

Definition:

Electromigration is the gradual displacement of metal atoms in interconnects due to high current density (J) over time.

Formula:

$$J = I/A$$

Where,

J = current density (A/cm²),

I = current,

A = cross-sectional area.

Effects:

- Creates voids (opens) or hillocks (shorts).
- Leads to wire failure or increased resistance.

➔ Causes of Electromigration

High current density (J)

- from narrow wires carrying high current

High temperature

- accelerates atomic diffusion.

Long metal lengths

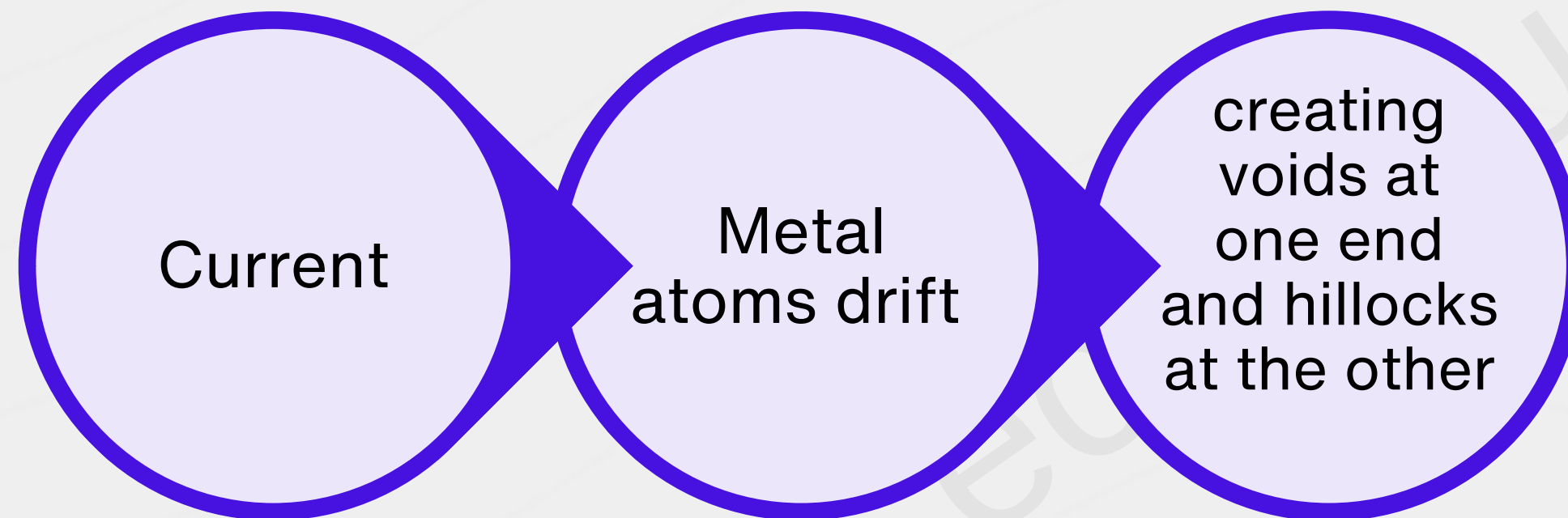
- more prone to stress migration

Poor PDN design

- local hotspots of current concentration

→ EM Visualization

Conceptual Diagram:



EM Visualization

Results:

- Voids → open circuits
- Hillocks → shorts with neighboring lines

Critical for:

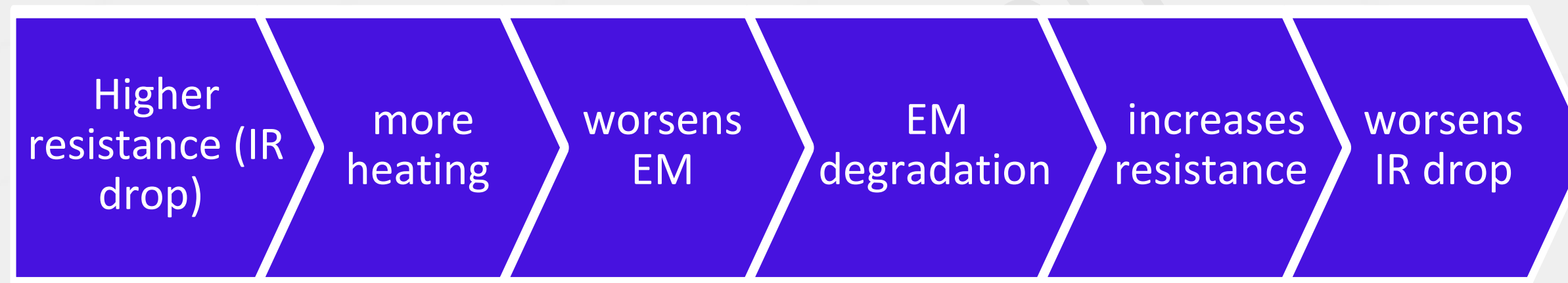
- Clock and power nets (carry continuous current).
- Wide metal layers used in power distribution.

Relation between IR Drop and EM

Aspect	IR Drop	Electromigration
Cause	Resistance $\times$ Current	High Current Density
Effect	Voltage drop	Metal degradation
Time Scale	Immediate (dynamic/static)	Long-term reliability
Prevention	Wider wires, denser power grid	Limit current density, cooling, redundant vias

➔ **Relation between IR Drop and EM**

Both issues are related —



➔ Example Flow (Conceptual)

While advanced IR/EM tools are commercial (e.g., RedHawk, Voltus), a conceptual open-source check can be done:

1. Extract parasitic resistances using OpenRCX:

```
openrcx -lef tech.lef -def design_routed.def -spdef design.spdef
```

➔ **Example Flow (Conceptual)**

2. Estimate voltage drop using current activity and resistance:

$$V_{\text{drop}} = I * R$$

3. Flag nets exceeding IR or EM limits based on design constraints.

➔ Mitigation Techniques

IR Drop Mitigation:

- Increase metal width on power rails.
- Add decoupling capacitors near switching blocks.
- Optimize cell placement to balance current demand.

EM Mitigation:

- Limit current per width ($J \leq J_{max}$).
- Use multiple vias for current sharing.
- Reduce operating temperature.
- Employ redundant paths for reliability

➔ Summary and Best Practices

Summary

IR Drop: Voltage drop due to resistance in power paths.

Electromigration: Atom movement due to current stress, reducing wire lifetime.

Both affect power integrity and chip reliability.

Best Practices

Design robust power grids early in layout planning

Run parasitic extraction to estimate R and IR drops.

Keep current density below threshold values in routing layers

Use redundant vias and wider metals on power nets.

Thank You

echiphu

